

Chapter 5

An optimum filter is a filter used for acquiring a best estimate of desired signal from noisy measurement.

- different from classic filters like LPE, HPE, RPE.
- they are ~~not~~ optimum because they are designed based on optimization theory:

to minimize the Mean Square Error (MSE) betⁿ a processed signal & a desired signal. or equivalently, provides best estimate of a desired signal from a measured signal.

Process of FIR filter design by window method:

designed to have frequency response

$$H_d(\omega)$$

desired unit sample response: $h_d(n)$

$$\text{or, } H_d(\omega) = \sum_{n=-\infty}^{\infty} h_d[n] e^{-j\omega n}$$

← F.T. →

i.e.

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

specify length of filter M

use a desired windowing function of length M as $w(n)$

now,

$$H(\omega) = FT [h_d[n] \cdot w[n]]$$

by convolution property,

$$H(\omega) = H_d(\omega) * W(\omega)$$

↓

filter's response gotten.

A 82 Bh

Remez algorithm : published by Remez in 1934

- highly efficient iterative procedure, used to determine the locations of extremal frequencies & consists of the following steps.

1. A set of initial values for extremal frequencies are either chosen or are available from completion of previous iteration.

2. The value of ϵ is then computed by the expression :

$$\epsilon = \frac{c_0 \tilde{H}_d(\omega_0) + c_1 \tilde{H}_d(\omega_1) + \dots + c_{L+1} \tilde{H}_d(\omega_{L+1})}{\frac{c_0}{\tilde{\omega}_0(\omega_0)} + \frac{c_1}{\tilde{\omega}_1(\omega_1)} + \frac{c_2}{\tilde{\omega}_2(\omega_2)} + \dots + \frac{(-1)^{L+1} c_{L+1}}{\tilde{\omega}_{L+1}(\omega_{L+1})}}$$

c_n is given by :

$$c_n = \frac{L+1}{\prod_{l=0, l \neq n}^{L+1} \frac{1}{\cos(\omega_n) - \cos(\omega_l)}}$$

3. Values of amplitude response $A(\omega)$ at $\omega = \omega_i$ are then computed using

$$A(\omega_i) = \frac{(-1)^i \epsilon}{\tilde{\omega}(\omega_i)} + \tilde{H}_d(\omega_i) \quad 0 \leq i \leq L+1$$

4. Polynomial $P(\omega)$ is determined by interpolating the $A(\omega_i)$ values at $L+2$ extremal frequencies using Lagrange interpolation formula

$$A(\omega) = \sum_{i=0}^{L+1} A(\omega_i) P_i(\cos \omega)$$

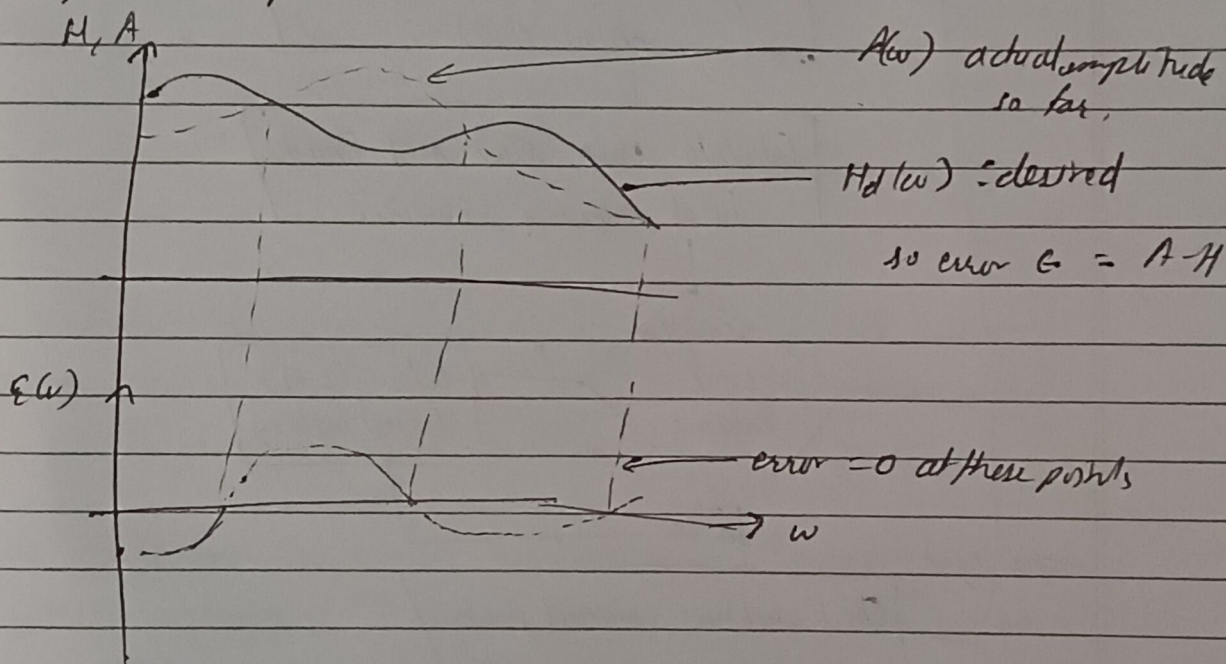
where,

$$P_i(\cos \omega) = \frac{1}{\prod_{l=0, l \neq i}^{L+1} \frac{\cos \omega - \cos \omega_l}{\cos \omega_i - \cos \omega_l}} \quad 0 \leq i \leq L+1$$

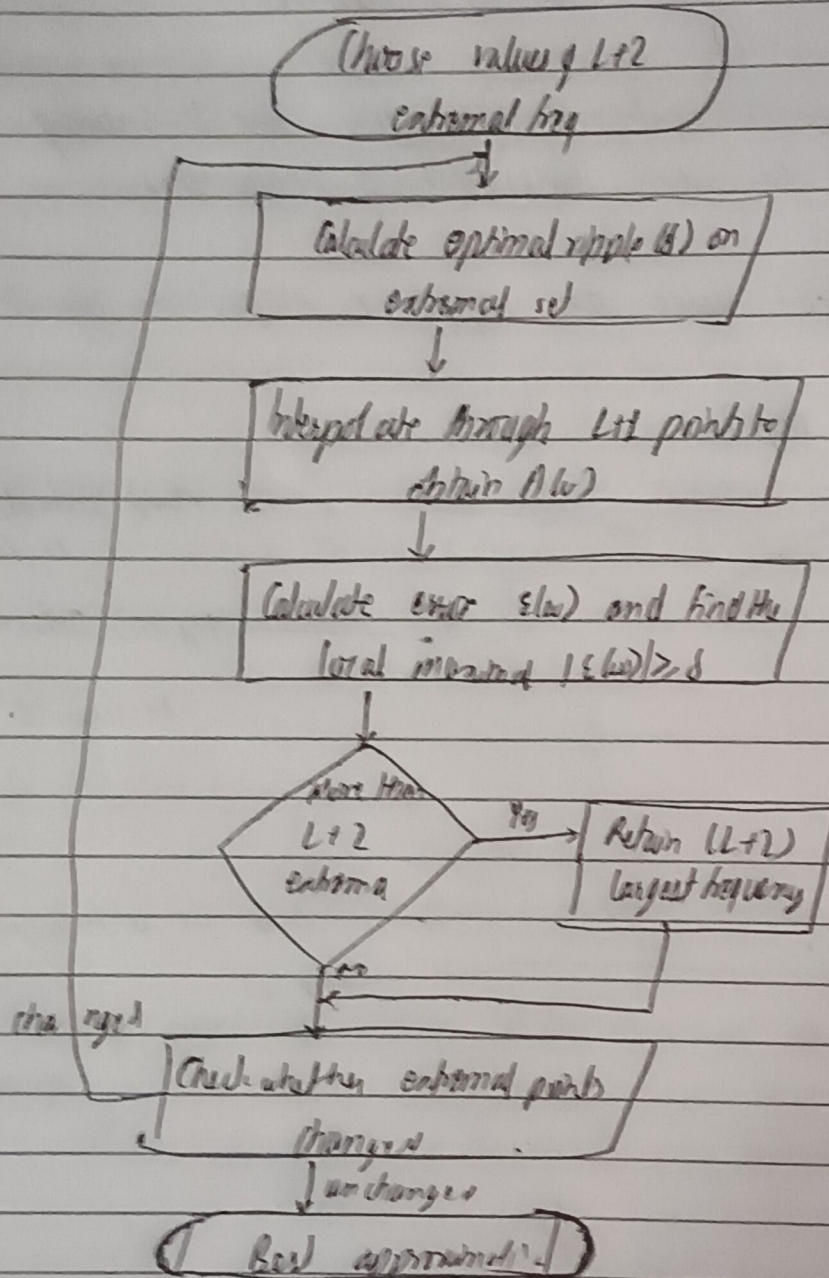
5. The new weighted error function $e(\omega)$ is computed at a dense set s ($s \geq L$) of frequencies.
In practice, $s = 16L$.

Determine $L+2$ new extremal frequencies from the values of $e(\omega)$ evaluated at the dense set of frequencies.

6. If $|e|$'s peak values are equal in magnitude, algorithm converged,
else go to step 2.



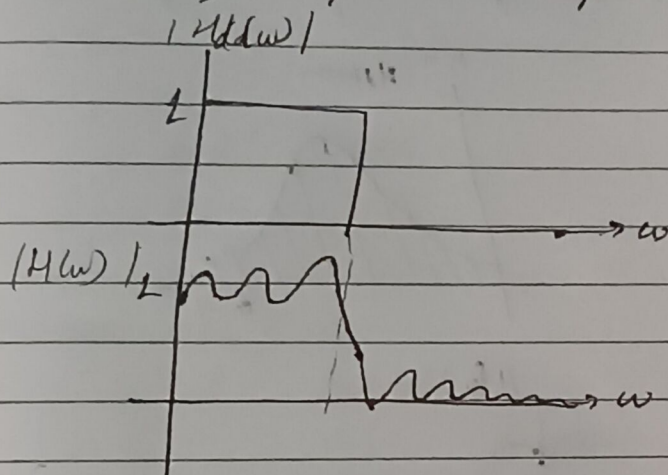
Flowchart



H 82 Ba

Gibbs Phenomenon

(desired)
for ideal vs expected (real/implementation)

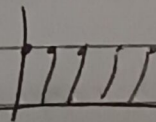


The oscillations produced at band edge (ω_c) of filter when applying window function is called Gibbs phenomenon, where:
 $H(\omega) = H_d(\omega) * W(\omega)$

These oscillations or ringing is generated because of sidelobe in frequency response $W(\omega)$ of window function. $W(\omega)$ is an approximation of some desired windowing, thus it leaks or absorbs some parts of spectrum, specially at corner frequency (band edge), so this oscillating behavior is called Gibbs's ph.

In case of rectangular windowing:

- generated by abrupt discontinuity : window is :

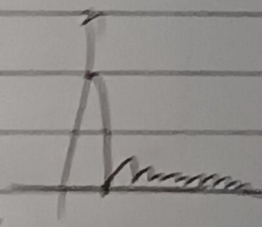
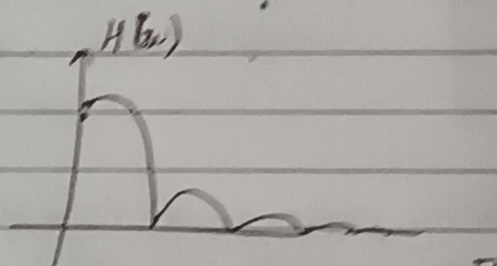
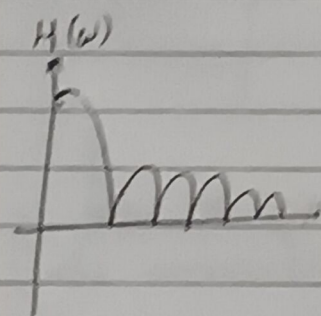
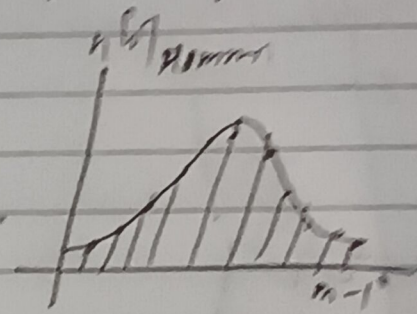
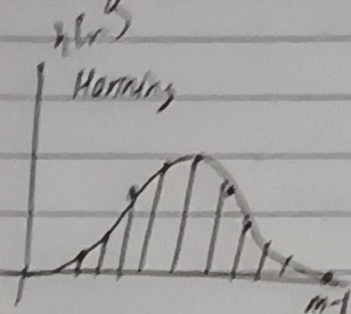


- sidelobe are larger in size since discontinuity is abrupt
- hence more effect in rectangular window.

H 81 B0

Gibb's Phenomena can be minimized by softening the response at band edge frequency.

Commonly used response like Hamming, Hanning have softer edges:



or using blackman for best ripple factor.

Kaiser window allows use of lengths M of filter separately than with width of main lobe / attenuation.

Part B: other windows like hamming / rectangular / blackman

controlled by 1 parameter alone $\rightarrow$ with M , difficult to have alternative.

use Kaiser, separate control of window & separate control of ripple / attenuation / main lobe.

But not independent, so good is $\star$

length $\uparrow$ = expensive $\uparrow$

control $\uparrow$

(2)

length not tied by attenuation (2)

#

80 Bn

Kaiser window:

$$|H(e^{j\omega})| \leq 0.01 \quad 0.21\pi \leq |\omega| \leq \pi, \text{ SB.}$$

$$0.95 \leq |H(e^{j\omega})| \leq 1.05, \quad 0.51\pi \leq 0.19\pi, \text{ PB.}$$

Here,

$$\delta_1 = 0.05 \text{ in PB}$$

$$\delta_2 = 0.01 \text{ in SB}$$

minimum value of ripple:

$$\delta = \min(\delta_1, \delta_2)$$

$$= 0.01 \leftarrow \text{optimum value of ripple}$$

$$\Delta\omega = \omega_s - \omega_p$$

$$= (0.21 - 0.19)\pi$$

$$= 0.02\pi$$

$$\text{Re } A = -20 \log_{10} \delta$$

$$= -20 \log_{10} 0.01$$

$$= 40 \quad (21.5 \text{ A } 550) \leftarrow \text{attenuation}$$

$$\omega_c = \frac{\omega_p + \omega_s}{2} = 0.20\pi$$

$$\beta = 0.5892 (A - 21)^{0.9} + 0.07886 (A - 21)$$

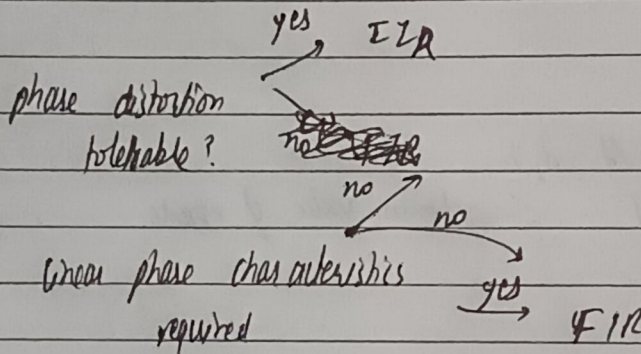
$$= 3.395$$

$$M, \text{ length} = \frac{A - 8}{2.285 \Delta\omega} = \frac{32}{2.285 \times 0.02\pi} = 222.88 \leftarrow \text{length}$$

$$\alpha = \frac{M}{2} = 111.5$$

It is 80 Bu

IIR filter has low no. of side lobes, but some phase distortion exist.
Linear phase characteristics are preserved by FIR filter.
if not required to preserve, any is ok.
do use case of both filter.



$$PB: 0.016\pi : \delta_1 = 0.01$$

$$SB: 0.08\pi : \delta_2 = 0.01$$

$$\delta = \min(\delta_1, \delta_2) = 0.01$$

$$\omega_c = 0.098\pi$$

$$\Delta\omega = 0.069\pi$$

$$A = -20 \log_{10} \delta$$

$$= 40 \quad (21 \leq A \leq 50)$$

$$\beta = 0.5842 (A - 21)^{0.4} + 0.07886 (A - 21)$$

$$= 3.395$$

$$M = \frac{A - 8}{2.185 \Delta\omega} = \frac{22.8}{2.185 \Delta\omega} = 69.65 \sim 70$$

$$d = \frac{M}{2} = 35$$

$$w(n) = \begin{cases} \frac{1}{3.395} \left[1 - \left(\frac{n-35}{35} \right)^2 \right]^{\frac{1}{2}} & 0 \leq n \leq 70 \\ 0 & \text{elsewhere} \end{cases} \quad 0.52223$$

$Z_0(x)$ can be calculated as:

$$Z_0(x) = 1 + \frac{0.25x^2}{(1!)^2} + \frac{(0.25x^2)^2}{(2!)^2} + \frac{(0.25x^2)^3}{(3!)^3} + \dots$$

For LPF filter, ideal desired frequency response is given by:

$$H_d(\omega) = \begin{cases} e^{-j\omega \left(\frac{M-1}{2} \right)} & \text{for } -\omega_c \leq \omega \leq \omega_c \\ 0 & \text{elsewhere} \end{cases}$$

↓ IFFT:

$$h_d[n] = \begin{cases} \frac{\sin \left[\omega_c \left(n - \frac{M-1}{2} \right) \right]}{\pi \left(n - \frac{M-1}{2} \right)} & \text{for } n \neq \frac{M-1}{2} \\ \omega_c / \pi & \text{for } n = \frac{M-1}{2} \end{cases}$$

$$\text{So, } h_d[n] = \begin{cases} \frac{\sin \left[0.048\pi \left(n - 34.5 \right) \right]}{\pi \left(n - 34.5 \right)} & \text{for } n \neq 34.5 \\ 0.048 & \text{for } n = 34.5 \end{cases}$$

not possible,

So, we increase length of filter by 1,

$$M_{\text{new}} = M + 1 = 71$$

$$h_d[n] = \begin{cases} \frac{\sin \left[0.048 \left(n - 35 \right) \right]}{\pi \left(n - 35 \right)} & \text{for } n \neq 35 \\ 0.048 & \text{for } n = 35 \end{cases}$$

so, finally :

$$h(n) = h_d(n) w(n)$$

$$= \begin{cases} \frac{\sin [0.048 \pi (n-35)]}{\pi (n-35)} \cdot \frac{70 \left[3.395 \left[1 - \left(\frac{n-35}{25} \right)^2 \right]^{\frac{1}{2}} \right]}{70 (3.395)} & \text{for } n \neq 35 \\ 0.098 & \text{for } n = 35 \end{cases}$$